

# Konstantinos Stamatis

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Materials scientist with over five years of experience leading projects, driven by a strong passion for contributing to innovative real-world applications and supporting strategic decision-making. Motivated by a continuous desire to excel in both science and leadership, I aim to combine my engineering expertise with in-depth knowledge of R&D processes to drive innovation and achieve business success.

## EDUCATION

Texas A&M University

Expected Graduation: December 2027

### Materials Science & Engineering, PhD

Focus on Polymeric Actuators for a Robotic Hand and Artificial Muscles

NCSR Demokritos

### Institute of Nanoscience & Nanotechnology, MS

Development of Metal Complexes for Anticancer Drugs

Aristotle University of Thessaloniki

### Chemistry Department, BS, Valedictorian

Major in Inorganic Chemistry

## RESEARCH EXPERIENCE

**NSF HAND Engineering Research Center (ERC) — Texas A&M University** - College Station, TX, USA

**August 2023- Present**

*Graduate Research Assistant, Dr. Taylor Ware and Dr. Svetlana Sukhishvili Laboratory,*

*Materials Science & Engineering and Biomedical Engineering Department*

- Conduct research within the NSF HAND ERC (\$52M, 10-year initiative), a 5-university consortium (Texas A&M, Northwestern, CMU, MIT, FAMU) aimed at developing next-generation dexterous robotic hands for intelligent, versatile robotic manipulation.
- Lead cross-laboratory collaboration to design and synthesize recyclable polymeric soft actuators – including **Liquid Crystal Elastomers (LCEs)** and **polydiethylsiloxane (PDES)** – for integration into AI-driven **robotic hand**.
- Perform comprehensive thermomechanical characterization of materials to evaluate and optimize actuation performance
- Leverage AI-assisted tools (Claude) to accelerate research progress, including Python script development for automated DMA/DSC/TGA data analysis and literature synthesis
- Drive commercialization efforts for advanced polymeric actuator materials, targeting applications in robotics and biomedical engineering.
- Mentor undergraduate researchers and serve as a Research Experiences for Undergraduates (REU) mentor.

**NCSR Demokritos – Athens, Greece**

**November 2021-August 2023**

*Graduate Research Assistant, Dr. Aikaterini Dendrinou-Samara, Dr. Aikaterini Raptopoulou & Dr. Vassilis Psycharis laboratory*

- Led a 1-year project across 3 labs and partnered with cross-functional teams to synthesize and characterize four zinc complexes, evaluating their antimicrobial and anticancer properties.
- Developed novel metal complexes demonstrating in vitro antimicrobial and anticancer activities superior to those of commercially available drugs.
- Speaker at the Joint CTMNM/NAGC Conference, May 2023, Spetses, Greece.

**NCSR Demokritos – Athens, Greece**

**September 2021-October 2021**

*Research Intern, Dr. Margarit Gjoka laboratory*

- Conducted research on advanced permanent magnet materials, with a focus on the synthesis and characterization of magnetic materials.
- Presented research findings at the 15th Joint MMM-INTERMAG Conference. January 2022, New Orleans, USA, and at the 1st Conference of NIP, February 2022, Tirana, Albania.

**Aristotle University of Thessaloniki – Thessaloniki, Greece**

**September 2017-July 2021**

*Undergraduate Research Assistant, Dr. Angaridis Panagiotis Laboratory*

- Selected from a competitive pool of students to conduct thesis research with a dissertation.
- Designed and synthesized novel copper complexes for OLED applications, conducting an in-depth study of OLED technology.
- Developed strong laboratory skills in synthesis and characterization techniques.

## PUBLICATIONS

- **Stamatis, K.**, Lee, Y. J., Dana, A., Kalairaj, M. S., Crusco, C. A., George, S. M., Sukhishvili, S. A., & Ware, T. H. *Thermomechanically Robust Liquid Crystal Elastomer Actuators with Diels–Alder Dynamic Bonds*. Under review at **Small** (Impact Factor 12.1).

## SCHOLARSHIPS AND AWARDS

**SPE (Society of Plastics Engineers) International Polyolefins Conference Scholarship**

**February 2026**

- Competitive graduate-level scholarship awarded for outstanding research excellence at Texas A&M University.

**Texas Soft Matter, Rice University**

**August 2025**

- Presentation Award Winner selected from 43 student participants for excellence in research presentation.

**Gerondelis Foundation Grant**

**July 2025**

- Awarded scholarship supporting Greek nationals in U.S. graduate programs

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| <b>Henry Kahn Memorial Scholarship</b><br><i>Society of Plastics Engineers (SPE) South Texas Section</i>                                                                                                                                                                                                                                                                                                                                                                                                   | <b>April 2025</b>                  |
| <ul style="list-style-type: none"> <li>Awarded for excellence in academics and research among graduate and undergraduate students across Texas.</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |                                    |
| <b>Material Advantage Congressional Visits Day (CVD) Participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>April 2025</b>                  |
| <ul style="list-style-type: none"> <li>Participated in the 2025 Material Advantage Congressional Visits Day in Washington, D.C.</li> <li>Awarded a \$320 travel grant to support participation in the event.</li> <li>Engaged with Texas Senators and Representatives to advocate for the support of education and research in the physical sciences and engineering, emphasizing their role in driving innovation and scientific progress.</li> </ul>                                                     |                                    |
| <b>Texas A&amp;M University</b><br><i>Aggies Invent: Problems Worth Solving Competition by the Entrepreneurship Department</i>                                                                                                                                                                                                                                                                                                                                                                             | <b>November 2024</b>               |
| <ul style="list-style-type: none"> <li>Addressing Costly and Life-Threatening Catheter-Associated UTIs</li> <li>First Place Award for Designing the Most Innovative Prototype</li> </ul>                                                                                                                                                                                                                                                                                                                   |                                    |
| <b>LEADERSHIP &amp; SERVICE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                    |
| <b>NSF HAND Engineering Research Center (ERC) — Texas A&amp;M University</b><br><i>Officer, Student Leadership Council (SLC)</i>                                                                                                                                                                                                                                                                                                                                                                           | <b>March 2026 - Present</b>        |
| <ul style="list-style-type: none"> <li>Spearhead professional development programming connecting NSF HAND graduate students with industry leaders in robotics and advanced manufacturing</li> </ul>                                                                                                                                                                                                                                                                                                        |                                    |
| <b>Texas A&amp;M University</b><br><i>President, Society of Plastics Engineers (SPE)</i>                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Summer 2025- Present</b>        |
| <ul style="list-style-type: none"> <li>Global professional organization advancing knowledge and education in the plastics industry</li> <li>Plan networking events with polymer industry professionals and coordinate site visits to major companies including Dow, Chevron Phillips, and Kuraray.</li> </ul>                                                                                                                                                                                              |                                    |
| <b>Texas A&amp;M University</b><br><i>Graduate Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Spring 2025 &amp; Fall 2025</b> |
| <ul style="list-style-type: none"> <li>Teach polymer labs to 80 undergraduate students at the Baker Hughes Materials Lab, Texas A&amp;M University.</li> <li>Supervise and mentor laboratory students in preparing detailed 30-page reports, executing advanced characterization and assay development protocols, and managing \$200K worth of lab equipment and materials.</li> </ul>                                                                                                                     |                                    |
| <b>Texas A&amp;M University</b><br><i>Vice President, Materials Science &amp; Engineering Graduate Student Advisory Council</i>                                                                                                                                                                                                                                                                                                                                                                            | <b>Fall 2023-Summer 2025</b>       |
| <ul style="list-style-type: none"> <li>Collaborated with Materials Science and Engineering department leadership to enhance student life and expand academic resources.</li> <li>Hosted town halls to engage graduate students, collect feedback, and successfully advocate for improvements that enhanced student experiences.</li> </ul>                                                                                                                                                                 |                                    |
| <b>Texas A&amp;M University</b><br><i>Founder &amp; President, Hellenic Student Association</i>                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Fall 2023-Present</b>           |
| <ul style="list-style-type: none"> <li>Founded, led, and secured official recognition for the organization at Texas A&amp;M University, inspiring student engagement, recruiting faculty support, and fostering active participation.</li> <li>Collaborated with Halliburton Engineering Global Programs leadership and Greek faculty to expand Texas A&amp;M's international student recruitment in Greece and Cyprus</li> </ul>                                                                          |                                    |
| <b>Organizations</b><br><i>Member, Material Advantage at Texas A&amp;M University</i><br><i>Member, Minerals, Metals &amp; Materials Society (TMS)</i><br><i>Member, American Society for Metals (ASM)</i><br><i>Member, American Ceramic Society (ACerS)</i><br><i>Member, Association for Iron &amp; Steel Technology (AIST)</i>                                                                                                                                                                         | <b>Spring 2025-Present</b>         |
| <b>Erasmus+</b><br><i>Greek Representative in Berlin, Germany, European Youth Exchange Program</i>                                                                                                                                                                                                                                                                                                                                                                                                         | <b>October 2018</b>                |
| <ul style="list-style-type: none"> <li>Selected and fully funded by the United Societies of the Balkans, a non-profit organization (NGO).</li> <li>Promoted youth mobility and collaborated with teams from five different countries: Bulgaria, Poland, Spain, Romania, and Italy.</li> <li>Addressed environmental awareness issues in European countries and explored renewable energy sources.</li> </ul>                                                                                               |                                    |
| <b>Aristotle University of Thessaloniki</b><br><i>Social Manager, Chemistry Professional Organization "ReAcTiON"</i>                                                                                                                                                                                                                                                                                                                                                                                       | <b>September 2018- August 2019</b> |
| <ul style="list-style-type: none"> <li>Organize events and outreach programs to promote the field of chemistry to the public.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |                                    |
| <b>TECHNICAL SKILLS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                    |
| <b>Material characterization:</b> Dynamic Mechanical Analysis (DMA), Universal Testing Machine (UTM), Differential Scanning Calorimetry (DSC), Gel Permeation Chromatography (GPC), Infrared (IR) Spectroscopy, Scanning Electron Microscopy (SEM), Thermogravimetric Analysis (TGA), Polarized Optical Microscopy (POM), Powder X-ray Diffraction (XRD), Vibrating Sample Magnetometry (VSM), Fluorescence Spectroscopy, H Nuclear Magnetic Resonance (NMR), UV/Vis Spectroscopy, Circular Dichroism (CD) |                                    |
| <b>Software:</b> Microsoft Office, JMP, SolidWorks, Origin, Python, AI-assisted workflows (Claude)                                                                                                                                                                                                                                                                                                                                                                                                         |                                    |
| <b>INTERESTS AND SKILLS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                    |
| <b>Languages:</b> Greek (Native), English (Fluent)                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                    |
| <b>Sports:</b> Competitive and Semi-Professional level in soccer                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                    |
| <b>Music:</b> piano, trumpet, guitar, and Byzantine music                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                    |